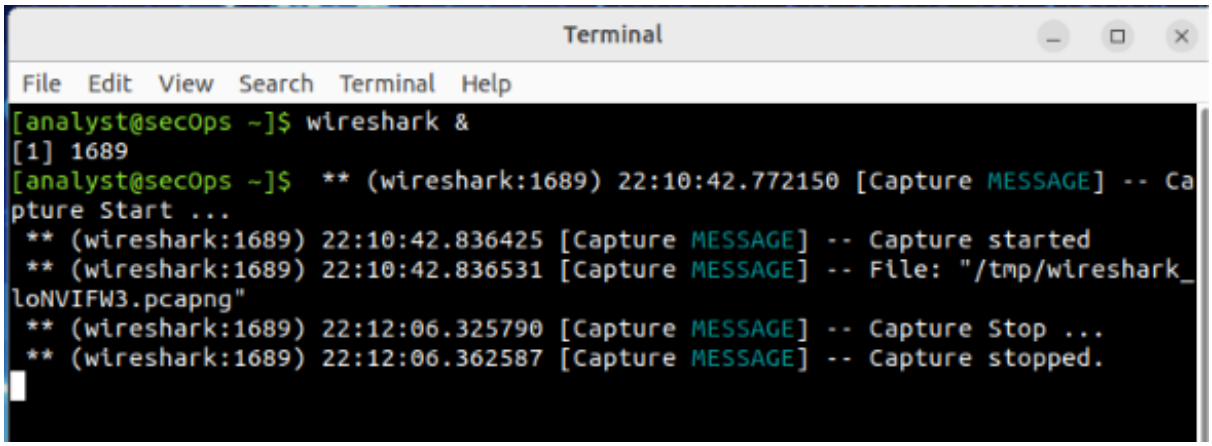
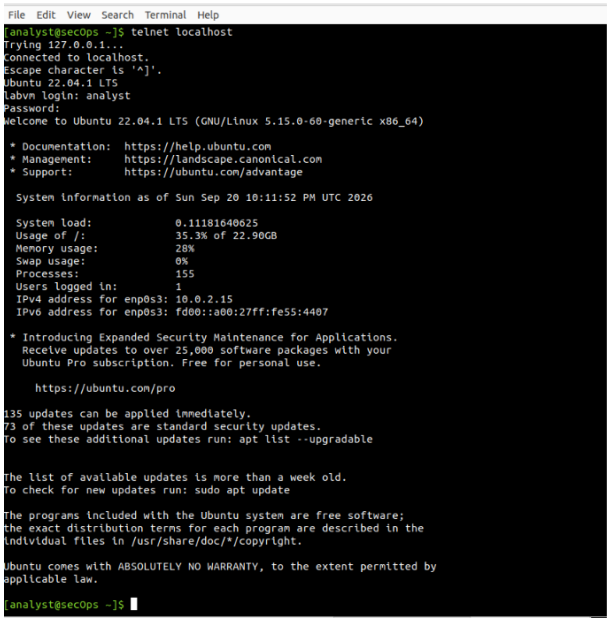
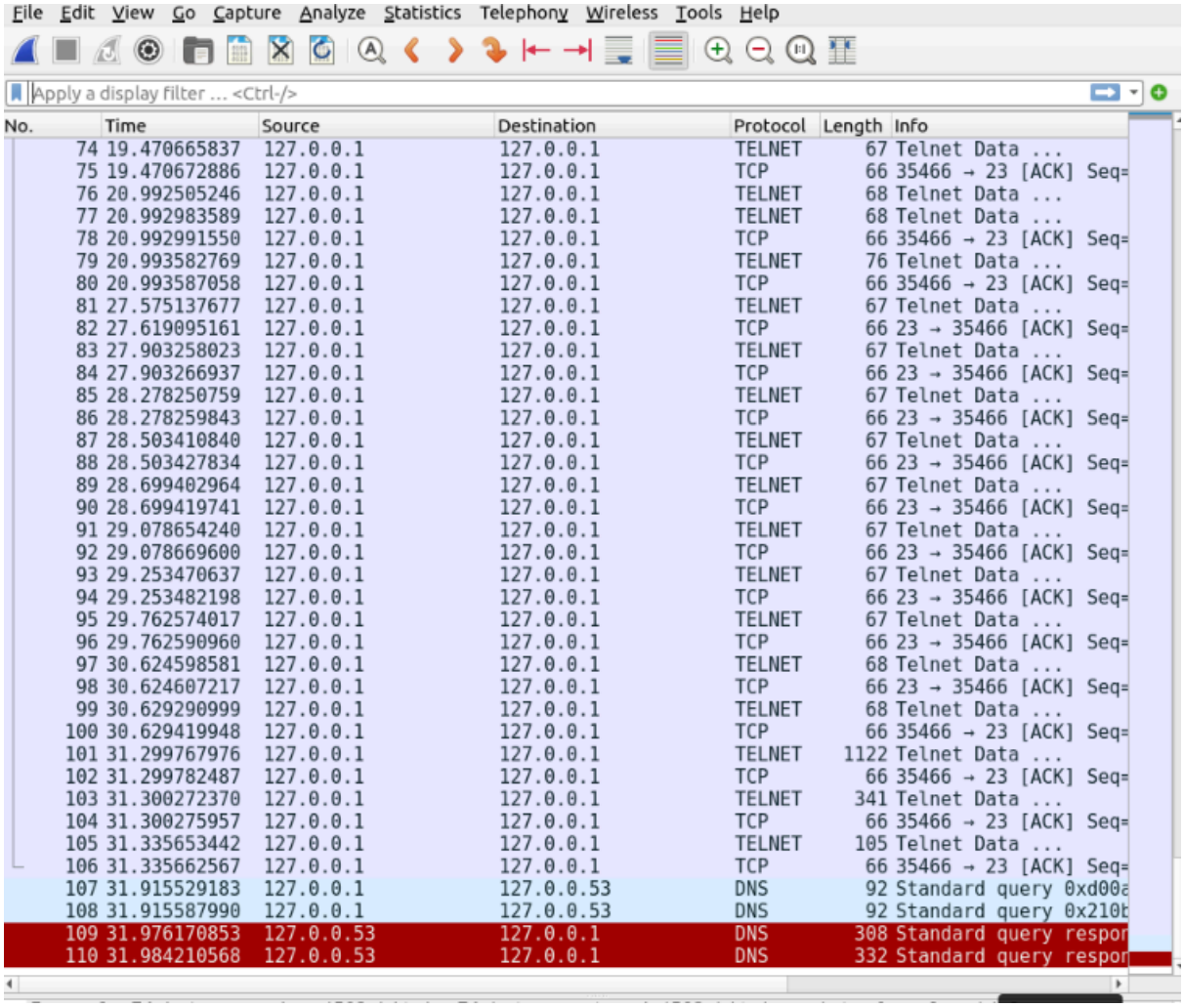
21.2.12-Lab - Examining Telnet and SSH in Wireshark

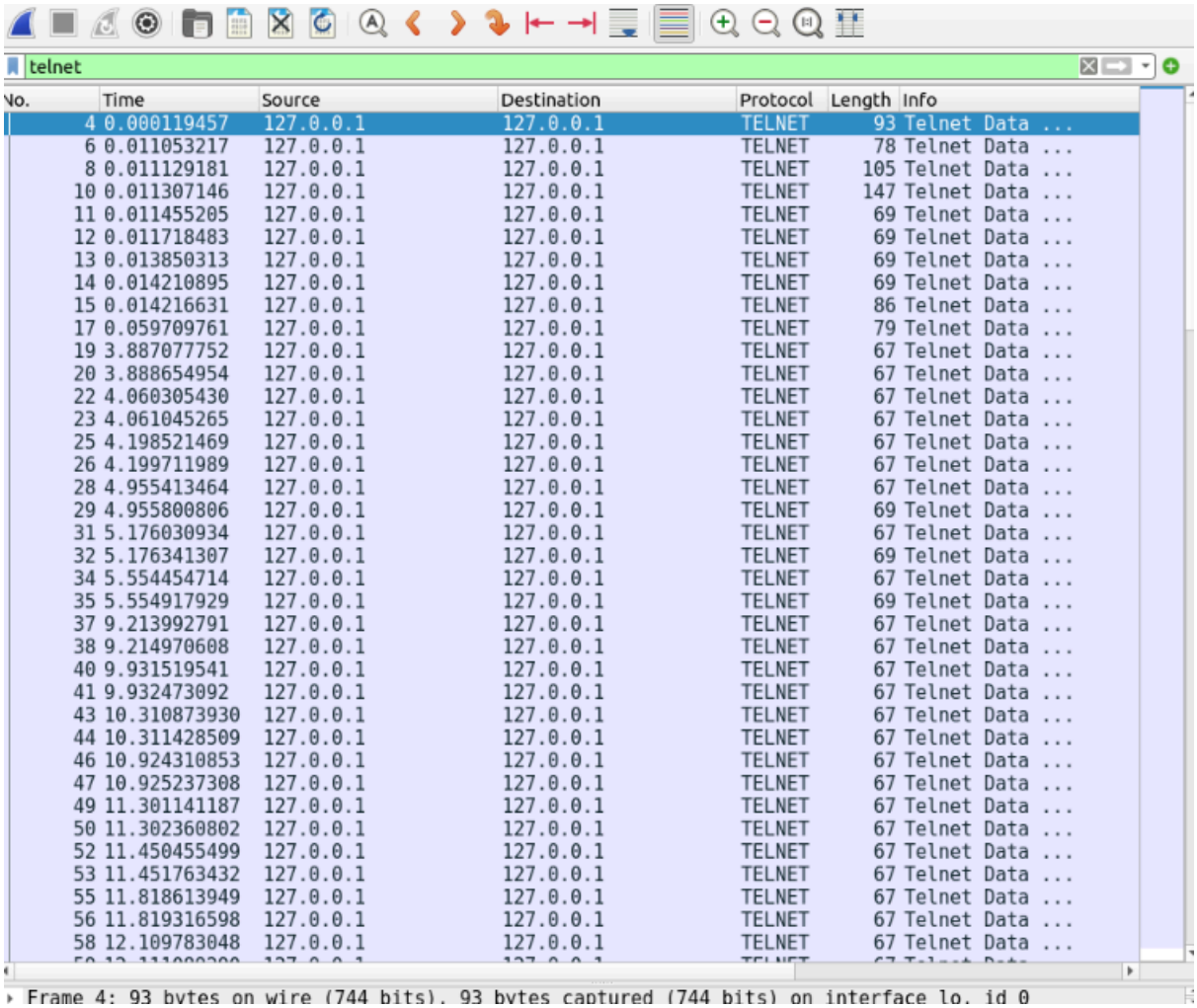
Ahmed Abdelhedi





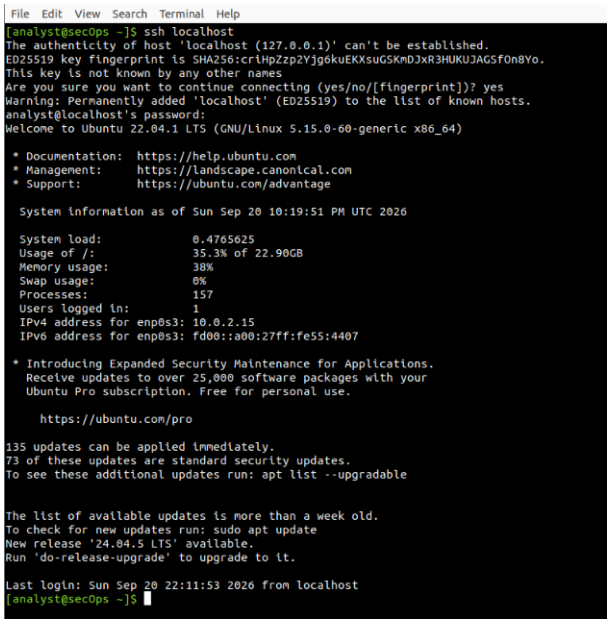

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
74 19.470665837 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
75 19.470672886 127.0.0.1 127.0.0.1 TCP 66 35466 → 23 [ACK] Seq=
76 20.992505246 127.0.0.1 127.0.0.1 TELNET 68 Telnet Data ...
77 20.992983589 127.0.0.1 127.0.0.1 TELNET 68 Telnet Data ...
78 20.992991550 127.0.0.1 127.0.0.1 TCP 66 35466 → 23 [ACK] Seq=
79 20.993582769 127.0.0.1 127.0.0.1 TELNET 76 Telnet Data ...
80 20.993587058 127.0.0.1 127.0.0.1 TCP 66 35466 → 23 [ACK] Seq=
81 27.575137677 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
82 27.619095161 127.0.0.1 127.0.0.1 TCP 66 23 → 35466 [ACK] Seq=
83 27.903258023 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
84 27.903266937 127.0.0.1 127.0.0.1 TCP 66 23 → 35466 [ACK] Seq=
85 28.278250759 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
86 28.278259843 127.0.0.1 127.0.0.1 TCP 66 23 → 35466 [ACK] Seq=
87 28.503410840 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
88 28.503427834 127.0.0.1 127.0.0.1 TCP 66 23 → 35466 [ACK] Seq=
89 28.699402964 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
90 28.699419741 127.0.0.1 127.0.0.1 TCP 66 23 → 35466 [ACK] Seq=
91 29.078654240 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
92 29.078669600 127.0.0.1 127.0.0.1 TCP 66 23 → 35466 [ACK] Seq=
93 29.253470637 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
94 29.253482198 127.0.0.1 127.0.0.1 TCP 66 23 → 35466 [ACK] Seq=
95 29.762574017 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
96 29.762590960 127.0.0.1 127.0.0.1 TCP 66 23 → 35466 [ACK] Seq=
97 30.624598581 127.0.0.1 127.0.0.1 TELNET 68 Telnet Data ...
98 30.624607217 127.0.0.1 127.0.0.1 TCP 66 23 → 35466 [ACK] Seq=
99 30.629290999 127.0.0.1 127.0.0.1 TELNET 68 Telnet Data ...
100 30.629419948 127.0.0.1 127.0.0.1 TCP 66 35466 → 23 [ACK] Seq=
101 31.299767976 127.0.0.1 127.0.0.1 TELNET 1122 Telnet Data ...
102 31.299782487 127.0.0.1 127.0.0.1 TCP 66 35466 → 23 [ACK] Seq=
103 31.300272370 127.0.0.1 127.0.0.1 TELNET 341 Telnet Data ...
104 31.300275957 127.0.0.1 127.0.0.1 TCP 66 35466 → 23 [ACK] Seq=
105 31.335653442 127.0.0.1 127.0.0.1 TELNET 105 Telnet Data ...
106 31.335662567 127.0.0.1 127.0.0.1 TCP 66 35466 → 23 [ACK] Seq=
107 31.915529183 127.0.0.1 127.0.0.53 DNS 92 Standard query 0xd00a
108 31.915587990 127.0.0.1 127.0.0.53 DNS 92 Standard query 0x210b
109 31.976170853 127.0.0.53 127.0.0.1 DNS 308 Standard query respor
110 31.984210568 127.0.0.53 127.0.0.1 DNS 332 Standard query respor

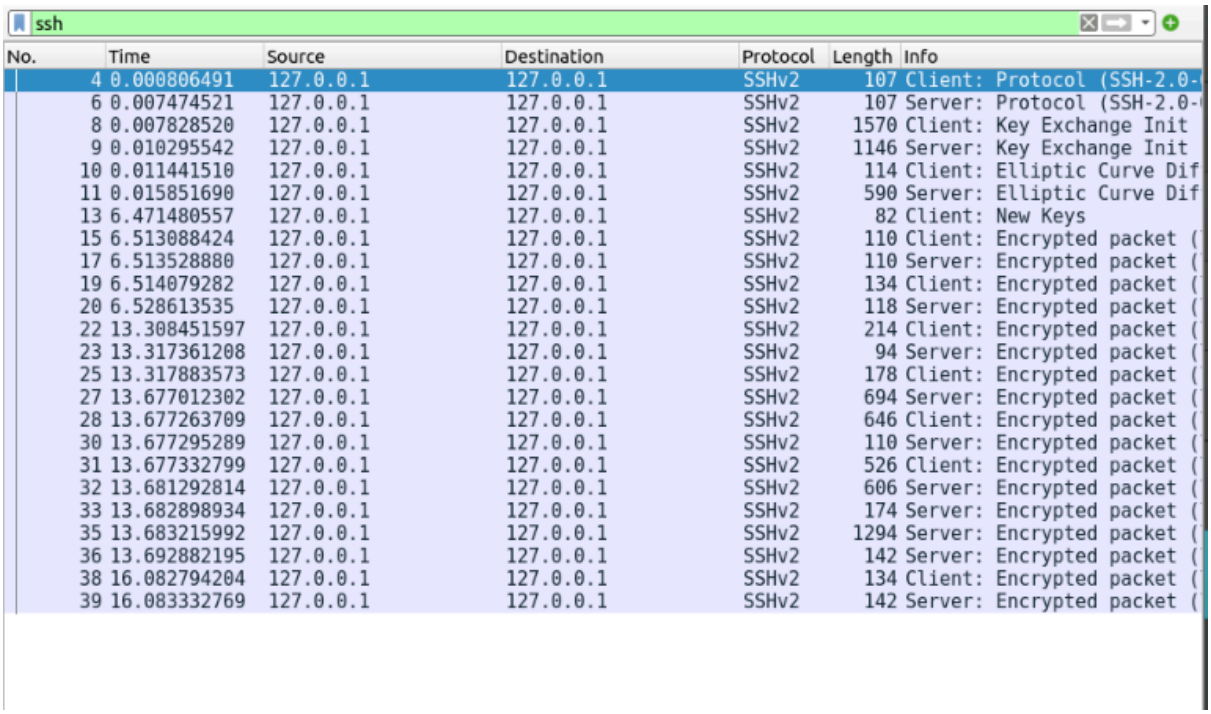
step2:

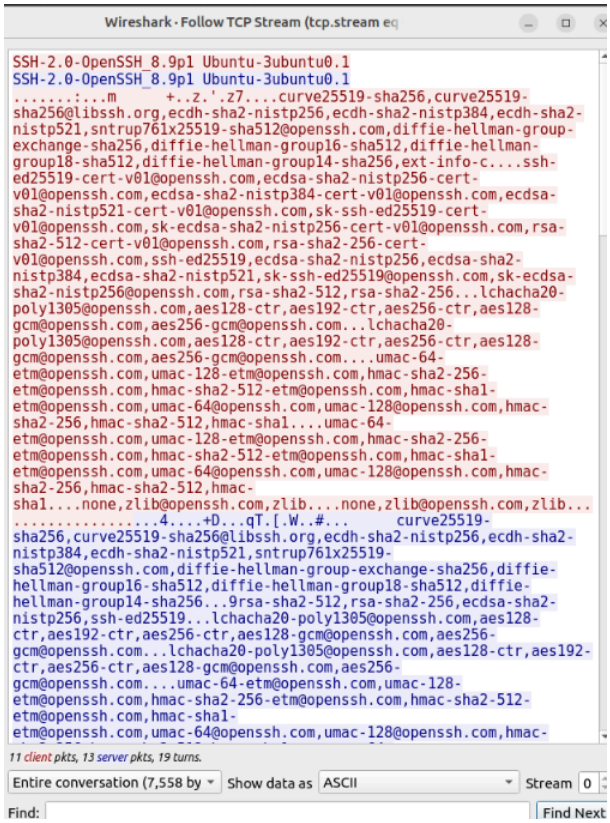

telnet
No. Time Source Destination Protocol Length Info
4 0.000119457 127.0.0.1 127.0.0.1 TELNET 93 Telnet Data ...
6 0.011053217 127.0.0.1 127.0.0.1 TELNET 78 Telnet Data ...
8 0.011129181 127.0.0.1 127.0.0.1 TELNET 105 Telnet Data ...
10 0.011307146 127.0.0.1 127.0.0.1 TELNET 147 Telnet Data ...
11 0.011455205 127.0.0.1 127.0.0.1 TELNET 69 Telnet Data ...
12 0.011718483 127.0.0.1 127.0.0.1 TELNET 69 Telnet Data ...
13 0.013850313 127.0.0.1 127.0.0.1 TELNET 69 Telnet Data ...
14 0.014210895 127.0.0.1 127.0.0.1 TELNET 69 Telnet Data ...
15 0.014216631 127.0.0.1 127.0.0.1 TELNET 86 Telnet Data ...
17 0.059709761 127.0.0.1 127.0.0.1 TELNET 79 Telnet Data ...
19 3.887077752 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
20 3.888654954 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
22 4.060305430 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
23 4.061045265 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
25 4.198521469 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
26 4.199711989 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
28 4.955413464 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
29 4.955800806 127.0.0.1 127.0.0.1 TELNET 69 Telnet Data ...
31 5.176030934 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
32 5.176341307 127.0.0.1 127.0.0.1 TELNET 69 Telnet Data ...
34 5.554454714 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
35 5.554917929 127.0.0.1 127.0.0.1 TELNET 69 Telnet Data ...
37 9.213992791 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
38 9.214970608 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
40 9.931519541 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
41 9.932473092 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
43 10.310873930 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
44 10.311428509 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
46 10.924310853 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
47 10.925237308 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
49 11.301141187 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
50 11.302360802 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
52 11.450455499 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
53 11.451763432 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
55 11.818613949 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
56 11.819316598 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
58 12.109783048 127.0.0.1 127.0.0.1 TELNET 67 Telnet Data ...
Frame 4: 93 bytes on wire (744 bits), 93 bytes captured (744 bits) on interface lo, id 0


Wireshark · Follow TCP Stream (tcp.stream eq
....!.."..'.....#..... ..#..'........!.."..... .....#.....'.....P..... .38400,38400....#.labvm: 0....'..DISPLAY.labvm:0.....xterm-256color.....Ubuntu 22.04.1 LTS
labvm login: qqnnqq.. ... ... .aannaallyyaattee.. ... ... .sstt
Password: cyberops
Welcome to Ubuntu 22.04.1 LTS (GNU/Linux 5.15.0-60-generic x86_64)
* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/advantage
System information as of Sun Sep 20 10:11:52 PM UTC 2026

part2:


File Edit View Search Terminal Help
[analyst@secOps ~]$ ssh localhost
The authenticity of host 'localhost (127.0.0.1)' can't be established.
ED25519 key fingerprint is SHA256:criHpZzp2Yjg6kuEKXsuGSKmDJxR3HUKUJAGSfOn8Yo.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
analyst@localhost's password:
Welcome to Ubuntu 22.04.1 LTS (GNU/Linux 5.15.0-60-generic x86_64)
* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/advantage
System information as of Sun Sep 20 10:19:51 PM UTC 2026
System load: 0.4765625
Usage of /: 35.3% of 22.90GB
Memory usage: 38%
Swap usage: 0%
Processes: 157
Users logged in: 1
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd00::a00:27ff:fe55:4407
* Introducing Expanded Security Maintenance for Applications.
Receive updates to over 25,000 software packages with your
Ubuntu Pro subscription. Free for personal use.
https://ubuntu.com/pro
135 updates can be applied immediately.
73 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable
The list of available updates is more than a week old.
To check for new updates run: sudo apt update
New release '24.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.
Last login: Sun Sep 20 22:11:53 2026 from localhost
[analyst@secOps ~]$


ssh
No. Time Source Destination Protocol Length Info
4 0.000806491 127.0.0.1 127.0.0.1 SSHv2 107 Client: Protocol (SSH-2.0-
6 0.007474521 127.0.0.1 127.0.0.1 SSHv2 107 Server: Protocol (SSH-2.0-
8 0.007828520 127.0.0.1 127.0.0.1 SSHv2 1570 Client: Key Exchange Init
9 0.010295542 127.0.0.1 127.0.0.1 SSHv2 1146 Server: Key Exchange Init
10 0.011441510 127.0.0.1 127.0.0.1 SSHv2 114 Client: Elliptic Curve Dif
11 0.015851690 127.0.0.1 127.0.0.1 SSHv2 590 Server: Elliptic Curve Dif
13 6.471480557 127.0.0.1 127.0.0.1 SSHv2 82 Client: New Keys
15 6.513088424 127.0.0.1 127.0.0.1 SSHv2 110 Client: Encrypted packet (
17 6.513528880 127.0.0.1 127.0.0.1 SSHv2 110 Server: Encrypted packet (
19 6.514079282 127.0.0.1 127.0.0.1 SSHv2 134 Client: Encrypted packet (
20 6.528613535 127.0.0.1 127.0.0.1 SSHv2 118 Server: Encrypted packet (
22 13.308451597 127.0.0.1 127.0.0.1 SSHv2 214 Client: Encrypted packet (
23 13.317361208 127.0.0.1 127.0.0.1 SSHv2 94 Server: Encrypted packet (
25 13.317883573 127.0.0.1 127.0.0.1 SSHv2 178 Client: Encrypted packet (
27 13.677012302 127.0.0.1 127.0.0.1 SSHv2 694 Server: Encrypted packet (
28 13.677263709 127.0.0.1 127.0.0.1 SSHv2 646 Client: Encrypted packet (
30 13.677295289 127.0.0.1 127.0.0.1 SSHv2 110 Server: Encrypted packet (
31 13.677332799 127.0.0.1 127.0.0.1 SSHv2 526 Client: Encrypted packet (
32 13.681292814 127.0.0.1 127.0.0.1 SSHv2 606 Server: Encrypted packet (
33 13.682898934 127.0.0.1 127.0.0.1 SSHv2 174 Server: Encrypted packet (
35 13.683215992 127.0.0.1 127.0.0.1 SSHv2 1294 Server: Encrypted packet (
36 13.692882195 127.0.0.1 127.0.0.1 SSHv2 142 Server: Encrypted packet (
38 16.082794204 127.0.0.1 127.0.0.1 SSHv2 134 Client: Encrypted packet (
39 16.083332769 127.0.0.1 127.0.0.1 SSHv2 142 Server: Encrypted packet (



SSH encrypts all data exchanged between the client and the server. Telnet, on the other hand, transmits information in plain text without encryption.

Reflection Question:

SSH is preferred because it encrypts the data exchanged during the remote connection, including usernames and passwords. Telnet sends data in plain text, making it much easier for attackers to intercept sensitive information.